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VOLTAGE DETECTION DEVICE AND BATTERY MODULE

Abstract

A front voltage detection device (**30**) includes a holder (**310**), a plurality of voltage detection lines (**320**), and a socket (**330**). The plurality of voltage detection lines (**320**) is electrically connected to a plurality of battery cells (**100**). The plurality of voltage detection lines (**320**) is held on the holder (**310**). The socket (**330**) is electrically connected to the plurality of voltage detection lines (**320**). The socket (**330**) is provided on the holder (**310**).

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a voltage detection device and a battery module.

BACKGROUND ART

[0002] For example, as described in Patent Document 1, a battery module includes a plurality of battery cells, a housing, a wire harness, and a connector. The housing accommodates the plurality of battery cells. The connector is electrically connected to the plurality of battery cells via the wire harness. In a voltage detection device described in Patent Document 1, the connector is attached to the housing.

RELATED DOCUMENT

Patent Document

[0003] Patent Document 1: Japanese Patent Application Publication (Translation of PCT Application) No. 2020-517051

SUMMARY OF INVENTION

Technical Problem

[0004] For example, as described in Patent Document 1, when a connector is attached to a housing, the connector is attached to an accommodation body after a plurality of battery cells and a wire harness are accommodated in the housing. In this case, however, for example, the necessity of an extra length of a wire harness for drawing out the wire harness to an outside of a housing may cause reducing workability for assembling a battery module.

[0005] One example of an object of the present invention is to improve workability for assembling a battery module. Another object of the present invention will be clarified from description of the present specification.

Solution to Problem

[0006] One aspect of the present invention is as follows. [0007] [1]

[0008] A voltage detection device including: [0009] a holder; [0010] a plurality of voltage detection lines electrically connected to a plurality of battery cells and held on the holder; and [0011] a connector electrically connected to the plurality of voltage detection lines and provided on the holder. [0012] [2]

[0013] The voltage detection device according to [1], in which the connector is provided with a terminal hole through which at least a portion of an external terminal is inserted. [0014] [3]

[0015] The voltage detection device according to [1] or [2], in which there is a gap through which at least a portion of an external connector to be connected to the connector is inserted. [0016] [4]

[0017] The voltage detection device according to any one of [1] to [3], further including attaching structure attaching the holder and the connector to each other. [0018] [5]

[0019] A voltage detection device including: [0020] a holder; [0021] a plurality of voltage detection lines electrically connected to a plurality of battery cells and held on the holder; and [0022] a socket electrically connected to the plurality of voltage detection lines, in which [0023] at least a portion of the socket is embedded in a recess portion provided in the holder. [0024] [6]

[0025] The voltage detection device according to [5], in which [0026] there is a gap between the socket and the recess portion. [0027] [7]

[0028] The voltage detection device according to [5] or [6], in which [0029] the plurality of battery cells is stacked in a predetermined first direction, [0030] the holder is located on a side of a second direction orthogonal to the first direction with respect to the plurality of battery cells, and [0031] the recess portion is provided on a surface of the holder on a side of a third direction orthogonal to the first direction and the second direction. [0032] [8]

[0033] A battery module including: [0034] the plurality of battery cells; and [0035] the voltage detection device according to any one of [1] to [7].

Advantageous Effects of Invention

[0036] Workability for assembling a battery module can be improved according to the above aspect of the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0037] FIG. **1** A front perspective view of a battery module according to an embodiment.
[0038] FIG. **2** A diagram with an accommodation body removed from FIG. **1**.
[0039] FIG. **3** An enlarged perspective view of one portion of a front voltage detection device according to the embodiment.
[0040] FIG. **4** A diagram with two sockets and a plug removed from FIG. **3**.
[0041] FIG. **5** A rear perspective view of the socket according to the embodiment.
[0042] FIG. **6** An enlarged perspective view of a front portion of a front voltage detection device according to a variant 1.
[0043] FIG. **7** An enlarged perspective view of a front portion of a front voltage detection device according to a variant 2.
[0044] FIG. **8** An enlarged perspective view of one portion of a front voltage detection device according to a variant 3.
DESCRIPTION OF EMBODIMENTS
[0045] Hereinafter, an embodiment according to the present invention is described by using the drawings. In all drawings, a similar constituent element is indicated by a similar reference sign, and description thereof is omitted as necessary.
[0046] FIG. **1** is a front perspective view of a battery module **1** according to an embodiment. FIG. **2** is a diagram with accommodation body **20** removed from FIG. **1**.
[0047] In each drawing, an arrow indicating an X direction, a Y direction, and a Z direction is illustrated for explanation. The X direction indicates a front-rear direction of the battery module **1**. Hereinafter, unless otherwise specifically noted, a tip end side of the arrow indicating the X direction is a rear side of the battery module **1**. Hereinafter, unless otherwise specifically noted, a proximal end side of the arrow indicating the X direction is a front side of the battery module **1**. The Y direction is orthogonal to the X direction. The Y direction indicates a left-right direction of the battery module **1**. Hereinafter, unless otherwise specifically noted, a tip end side of the arrow indicating the Y direction is a left side of the battery module **1**. Hereinafter, unless otherwise specifically noted, a proximal end side of the arrow indicating the Y direction is a right side of the battery module **1**. The Z direction is orthogonal to both of the X direction and the Y direction. The Z direction indicates an up-down direction of the battery module **1**. Hereinafter, unless otherwise specifically noted, a tip end side of the arrow indicating the Z direction is an upper side of the battery module **1**. Hereinafter, unless otherwise specifically noted, a proximal end side of the arrow indicating the Z direction is a lower side of the battery module **1**. The relations between the X direction, the Y direction, and the Z direction, and the front-rear direction, the left-right direction, and the up-down direction of the battery module **1**, however, is not limited to the above example. These relations differ according to an actual arrangement of the battery module **1**.
[0048] The battery module **1** includes a cell stack **10**, the accommodation body **20**, a front voltage detection device **30**, and a rear voltage detection device **30'**. The cell stack **10** includes a plurality of cell groups **100**G. Each cell group **100**G includes a plurality of battery cells **100**. The accommodation body **20** includes a front cover **210**, a rear cover **220**, a right cover **230**, a left cover **240**, a lower cover **250**, and an upper cover **260**. The front voltage detection device **30** includes a holder **310**, a plurality of voltage detection lines **320**, two sockets **330**, and a terminal **340**. A plurality of voltage detection tip end portions **322** are provided at ends of the plurality of voltage detection lines **320** on a side of a tab group **110** described later.

[0049] The plurality of battery cells **100** are stacked in the Y direction. Each battery cell **100** has a substantially rectangular shape when viewed from the Y direction. A long direction of each battery cell **100** is substantially in parallel to the X direction. A short direction of each battery cell **100** is substantially in parallel to the Z direction. A thickness direction of each battery cell **100** is substantially in parallel to the Y direction. The structure of each battery cell **100**, however, is not limited to this example.

[0050] Each battery cell **100** includes an unillustrated battery element, an exterior member **102**, a positive electrode tab **112**, and a negative electrode tab **114**. The exterior member **102** seals the battery element and unillustrated electrolytic liquid. The battery element includes a plurality of unillustrated positive electrodes and a plurality of unillustrated negative electrodes alternately stacked in the Y direction and an unillustrated separator located between the positive electrode and the negative electrode adjacent to each other in the Y direction. The positive electrode tab **112** and the negative electrode tab **114** are drawn out from opposite sides the exterior member **102** in the X direction. The positive electrode tab **112** is electrically connected to the above plurality of positive electrodes. The positive electrode tab **112** is composed of, for example, aluminum. The negative electrode tab **114** is electrically connected to the above plurality of negative electrodes. The negative electrode tab **114** is composed of, for example, copper.

[0051] Each cell group **100**G includes two battery cells **100** connected in parallel to each other. The positive electrode tabs **112** drawn out from two battery cells **100** included in each cell group **100**G face the same side in the X direction. The negative electrode tabs **114** drawn out from two battery cells **100** included in each cell group **100**G face the same side in the X direction. The positive electrode tabs **112** and the negative electrode tabs **114** drawn out from one of the cell groups **100**G adjacent to each other in the Y direction and the positive electrode tabs **112** and the negative electrode tabs **114** drawn from the other of the cell groups **100**G adjacent to each other in the Y direction face opposite sides in the X direction.

[0052] The plurality of cell groups **100**G are connected in series. The cell groups **100**G adjacent to each other in the Y direction include a tab group **110** located on the front side or the rear side of the cell groups **100**G. The tab group **110** includes the two positive electrode tabs **112** drawn from one of the cell groups **100**G adjacent to each other in the Y direction and the two negative electrode tabs **114** drawn out from the other of the cell groups **100**G adjacent to each other in the Y direction. In the tab group **110**, the two positive electrode tabs **112** and the two negative electrode tabs **114** are joined to one another, for example, by laser welding. A plurality of tab groups **110** located on the front side of the cell stack **10** and a plurality of tab groups **110** located on the rear side of the cell stack **10** are alternately placed in the Y direction. In an example illustrated in FIG. **2**, therefore, the cell group **100**G located on a leftmost side and the cell group **100**G located second from the left side, for example, include the tab group **110** located on the front side of these cell groups **100**G. The cell group **100**G located second from the left side and the cell group **100**G located third from the left side include the tab group **110** located on the rear side of these cell groups **100**G.

[0053] The structure of the cell stack **10** is not limited to the above example. For example, each cell group **100**G may include three or more battery cells **100**. A plurality of single battery cells **100** may be connected in series.

[0054] The accommodation body **20** includes the cell stack **10**, the front voltage detection device **30**, and the rear voltage detection device **30′**. The front cover **210** covers the front side of the cell stack **10** and the front voltage detection device **30**. The rear cover **220** covers the rear side of the cell stack **10** and the rear voltage detection device **30′**. The right cover **230** covers a right side of the cell stack **10**. The left cover **240** covers a left side of the cell stack **10**. The lower cover **250** covers a lower side of the cell stack **10**. The upper cover **260** covers an upper side of the cell stack **10**.

[0055] The holder **310** is provided on the front side of the cell stack **10**. The holder **310** is formed of, for example, resin. The holder **310** integrally holds the plurality of voltage detection lines **320**, the plurality of voltage detection tip end portions **322**, and the two sockets **330**. Therefore, positioning the holder **310** with respect to the cell stack **10** enables to position the plurality of voltage detection lines **320** and the plurality of voltage detection tip end portions **322**. For example, the holder **310** is positioned with respect to the cell stack **10** by attaching the holder **310** to the accommodation body **20**. When the holder **310** is positioned, each of the plurality of voltage detection tip end portions **322** is located on the front side of each of the plurality of tab groups **110** located on the front side of the cell stack **10**.

[0056] The plurality of voltage detection lines **320** are, for example, a wire harness. Each of the plurality of voltage detection lines **320** electrically connects each of the plurality of tab groups **110** and the socket **330** to each other.

[0057] Specifically, one end of each of the plurality of voltage detection lines **320** is electrically connected to each of the plurality of tab groups **110** located on the front side of the cell stack **10** via each of the plurality of voltage detection tip end potions **322**. Each voltage detection tip end portion **322** is, for example, a conductive plate such as a metal plate. Each voltage detection tip end portion **322** is joined to a front surface of each tab group **110**, for example, by laser welding. The other end of each of the plurality of voltage detection lines **320** is electrically connected to the socket **330**.

[0058] The terminal **340** is electrically connected to the positive electrode tab **112** drawn out to the front side from the battery cell **100** disposed on the rightmost side. The cell stack **10** is electrically connectable to external equipment via the terminal **340**.

[0059] Similarly to the front voltage detection device **30**, the rear voltage detection device **30′** includes an unillustrated holder, a plurality of unillustrated voltage detection lines, an unillustrated connector, and an unillustrated terminal. The terminal of the rear voltage detection device **30′** is electrically connected to the negative electrode tab **114** drawn out to the rear side from the battery cell **100** disposed on the leftmost side.

[0060] FIG. **3** is an enlarged perspective view of one portion of the front voltage detection device **30** according to the embodiment. FIG. **4** is a diagram with the two sockets **330** and a plug **430** removed from FIG. **3**. FIG. **5** is a rear perspective view of the socket **330** according to the embodiment.

[0061] The two sockets **330** and the plug **430** are described with reference to FIG. **3**.

[0062] As illustrated in FIG. **3**, a step **262** is provided on a front end of the upper cover **260**. The step **262** makes the front end portion of the upper cover **260** located at the position lower in the Z direction than the rear portion of the front end portion of the upper cover **260**. The two sockets **330** illustrated in FIG. **3** are disposed in a space formed by the step **262**. This can improve volumetric efficiency of the battery module **1**.

[0063] The plug **430** illustrated in FIG. **3** is connectable to the left socket **330** illustrated in FIG. **3**. The plug **430** includes a plurality of unillustrated terminals and a plug cover **434**. The plurality of terminals of the plug **430** protrude toward the right side. The plug cover **434** surrounds the plurality of terminals of the plug **430** when viewed from a right side of the plug **430**. Accordingly, foreign matter such as a finger of a worker or dust can be less likely to contact with the plurality of terminals of the plug **430** as compared with when the plurality of terminals of the plug **430** are not surrounded by the plug cover **434**.

[0064] The left socket **330** illustrated in FIG. **3** includes a body **332** and a socket cover **334**. Some voltage detection lines **320** of the plurality of voltage detection lines **320** are electrically connected to the left socket **330** illustrated in FIG. **3**. In an example illustrated in FIG. **3**, the some the voltage detection lines **320** are connected to the socket **330** through a tube **324** attached to the socket **330**.

[0065] A plurality of terminal holes **332***a* is formed in the body **332**. A terminal is embedded inside the plurality of terminal holes **332***a*. In the socket **330** according to the embodiment, therefore, the terminal can be less likely to contact with foreign matter such as a finger of a worker or dust as compared with when the terminal of the socket **330** protrudes to outside. Each of the above plurality of terminals of the plug **430** is inserted through each of the plurality of terminal holes **332***a* in the Y direction. The socket **330** and the plug **430** are electrically connected to each other by inserting the plurality of terminals of the plug **430** through each of the plurality of terminal holes **332***a* in the Y direction.

[0066] The socket cover **334** surrounds at least a portion of the body **332** with a gap **336** between them. The socket **330** and the plug **430** are connected to each other by inserting at least a portion of the plug cover **434** through the gap **336** in the Y direction. Accordingly, the mechanical connection of the socket **330** and the plug **430** can be strengthened as compared with when the gap **336** and the plug cover **434** are not provided. The gap **336** and the plug cover **434**, however, may not be provided. Also in this case, the socket **330** and the plug **430** can be mechanically connected to each other by inserting each of the above plurality of terminals of the plug **430** through each of the plurality of terminal holes **332***a* in the Y direction.

[0067] The right socket **330** illustrated in FIG. **3** includes a configuration similar to that of the left socket **330** illustrated in FIG. **3**. When the two sockets **330** are provided, some of the voltage detection lines **320** can be electrically connected to one of the sockets **330** and the other of the voltage detection lines **320** can be electrically connected to the other socket **330**. For this reason, the number of terminal holes **332** of each socket **330** can be reduced as compared with when all the plurality of voltage detection lines **320** are electrically connected to a single socket **330**. For example, when at least some of the terminal holes **332***a* of each socket **330** are arranged in the Z direction, a width of each socket **330** in the Z direction can be reduced. This can prevent an upper surface of each socket **330** from protruding upward from an upper surface of the upper cover **260**. However, the front voltage detection device **30** may include only one socket **330** or may include three or more sockets **330**.

[0068] The right socket **330** illustrated in FIG. **3** is disposed on the front side with respect to the left socket **330** illustrated in FIG. **3**. That is, the two sockets **330** illustrated in FIG. **3** are displaced from each other in the X direction. Accordingly, a space for passing the plurality of voltage detection lines **320** connected to the right socket **330** illustrated in FIG. **3** can be secured on the rear side of the left socket **330** illustrated in FIG. **3**. Accordingly, a space in the Y direction for arranging the two sockets **330** illustrated in FIG. **3** can be reduced as compared with when the two sockets **330** illustrated in FIG. **3** are aligned in the X direction. The layout of the sockets **330**, however, is not limited to this example. For example, the two sockets **330** illustrated in FIG. **3** may be aligned in the X direction.

[0069] A method of attaching the left socket **330** illustrated in FIG. **3** to the holder **310** is described by using FIGS. **4** and **5**.

[0070] As illustrated in FIG. **4**, a pair of locking grooves **352** are formed on a front surface of the holder **310**. The pair of locking grooves **352** extend substantially in parallel to each other in the Y direction. As illustrated in FIG. **5**, a pair of locking portions **354** are provided on a rear surface of the socket cover **334**. The pair of locking portions **354** extend substantially in parallel to the Y direction. The pair of locking portions **354** are insertable through the pair of locking grooves **352** from a left side of the pair of locking grooves **352** in the Y direction. The socket **330** is attached to the holder **310** by inserting the pair of locking portions **354** through the pair of locking grooves **352** from a left side of the pair of locking grooves **352** in the Y direction. That is, the pair of locking grooves **352** and the pair of locking portions **354** are attaching structure attaching the holder **310** and the left socket **330** illustrated in FIG. **3** to each other.

[0071] The attaching structure, however, is not limited to an example of the pair of locking grooves **352** and the pair of locking portions **354**. The right socket **330** illustrated in FIG. **3** is also attached to the holder **310** by attaching structure similar to the attaching structure of the left socket **330** illustrated in FIG. **3**.

[0072] As described above, in an example illustrated in FIGS. **4** and **5**, the above attaching structure, that is, the pair of locking grooves **352** and the pair of locking portions **354** are located on a rear surface side of the socket **330**. Accordingly, the attaching structure does not interfere with the plug cover **434** when the plug cover **434** is inserted through the gap **336** in the Y direction.

Accordingly, the plug cover **434** does not need provided with a cutout for avoiding interference of the attaching structure. Accordingly, lowering of strength of the plug cover **434** due to the cutout can be suppressed. However, the plug cover **434** may be provided with the cutout according to needs.

[0073] In the embodiment, the socket **330** is provided on the holder **310**. Accordingly, the cell stack **10** and the front voltage detection device **30** can be accommodated in the accommodation body **20** after the front voltage detection device **30** is attached to the cell stack **10**. For example, when the socket **330** is provided on the accommodation body **20**, the socket **330** is provided on the accommodation body **20** after the cell stack **10** is accommodated in the accommodation body **20**. This case, however, requires an extra length of the plurality of voltage detection lines **320** for drawing out the plurality of voltage detection lines **320** to the outside of the accommodation body **20**. The embodiment, on the other hand, does not need consideration of the extra length of the plurality of voltage detection lines **320**. In the embodiment, the workability for assembling the battery module **1** can be therefore improved as compared with when the socket **330** is provided on the accommodation body **20**.

[0074] In the embodiment, the socket **330** serves as a connector provided on the front voltage detection device **30**, and the plug **430** serves as an external connector of the front voltage detection device **30**. However, a connector provided on the front voltage detection device **30** may serve as a plug, and an external connector of the front voltage detection device **30** may serve as a socket. This example also does not require consideration of the above extra length of the plurality of voltage detection lines **320** in assembling the battery module **1**. In the embodiment, the workability for assembling the battery module **1** can be therefore improved as compared with when the above plug is provided on the accommodation body **20**.

[0075] FIG. **6** is an enlarged perspective view of a front portion of a front voltage detection device **30**A according to a variant 1. The front voltage detection device **30**A according to the variant 1 is similar to the front voltage detection device **30** according to the embodiment except for the following point.

[0076] At least a portion of a socket **330**A is embedded in a recess portion **312**A formed on an upper surface of a holder **310**A. A gap **336**A is provided between an outer lateral surface of the socket **330**A and an inner lateral surface of the recess portion **312**A. A plug **430**A is connectable to the socket **330**A from an upper side of the socket **330**A. The socket **330**A and the plug **430**A are mechanically connected to each other by inserting a plug cover **434**A through the gap **336**A in the Z direction. Also in the variant 1, the mechanical connection of the socket **330**A and the plug **430**A can be therefore strengthened similarly to the embodiment.

[0077] FIG. **7** is an enlarged perspective view of a front portion of a front voltage detection device **30**B according to a variant 2. The front voltage detection device **30**B according to the variant 2 is similar to the front voltage detection device **30** according to the embodiment except for the following point.

[0078] At least a portion of a socket **330**B is embedded in a recess portion **312**B formed on a front surface of a holder **310**B. A gap **336**B is provided between an outer lateral surface of the socket **330**B and an inner lateral surface of the recess portion **312**B. A plug **430**B is connectable to the socket **330**B from the front side of the socket **330**B. The socket **330**B and the plug **430**B are mechanically connected to each other by inserting a plug cover **434**B through the gap **336**B in the X direction. Also in the variant 2, the mechanical connection of the socket **330**B and the plug **430**B can be therefore strengthened similarly to the embodiment.

[0079] FIG. **8** is an enlarged perspective view of one portion of a front voltage detection device **30**C according to a variant 3. The front voltage detection device **30**C according to the variant 3 is similar to the front voltage detection device **30** according to the embodiment except for the following point.

[0080] The front voltage detection device **30**C according to the variant 3 includes a socket holder

350C. The socket holder **350**C is attached to an unillustrated holder **310**. The socket holder **350**C holds a socket **330**C. In an example illustrated in FIG. **8**, a plurality of terminal holes **332***a*C is provided on a left lateral surface of the socket **330**C. The socket holder **350**C holds the socket **330**C with a gap **336**C formed between the socket **330**C and a portion of the socket holder **350**C surrounding the socket **330**C. The socket **330**C and an unillustrated plug **430** are mechanically connected to each other by inserting an unillustrated plug cover **434** through the gap **336**C in the Y direction similarly to the embodiment.

[0081] In the foregoing, an embodiment according to the present invention has been described with reference to the drawings; however, these are examples of the present invention, and various configurations other than the above can also be adopted.

[0082] This application claims priority based on Japanese patent application No. 2022-074766, filed on Apr. 28, 2022, the disclosure of which is incorporated herein in its entirety by reference.

REFERENCE SIGNS LIST

[0083] **1** Battery module, **10** Cell stack, **20** Accommodation body, **30**, **30**A, **30**B, **30**C Front voltage detection device, **100** Battery cell, **100**G Cell group, **102** Exterior member, **110** Tab group, **112** Positive electrode tab, **114** Negative electrode tab, **210** Front cover, **220** Rear cover, **230** Right cover, **240** Left cover, **250** Lower cover, **260** Upper cover, **262** Step, **310**, **310**A, **310**B Holder, **312**A, **312**B Recess portion, **320** Voltage detection line, **322** Voltage detection tip end portion, **324** Tube, **330**, **330**A, **330**B, **330**C Socket, **332** Body, **332***a*, **332***a*C Terminal hole, **334** Socket cover, **336**, **336**A, **336**B, **336**C Gap, **340** Terminal, **350**C Socket holder, **352** Locking groove, **354** Locking portion, **430**, **430**A, **430**B Plug, **434**, **434**A, **434**B Plug cover

Claims

1. A voltage detection device comprising: a holder; a plurality of voltage detection lines electrically connected to a plurality of battery cells and held on the holder; and a connector electrically connected to the plurality of voltage detection lines and provided on the holder.

2. The voltage detection device according to claim 1, wherein the connector is provided with a terminal hole through which at least a portion of an external terminal is inserted.

3. The voltage detection device according to claim 1, wherein there is a gap through which at least a portion of an external connector to be connected to the connector is inserted.

4. The voltage detection device according to claim 1, further comprising attaching structure attaching the holder and the connector to each other.

5. A voltage detection device comprising: a holder; a plurality of voltage detection lines electrically connected to a plurality of battery cells and held on the holder; and a socket electrically connected to the plurality of voltage detection lines, wherein at least a portion of the socket is embedded in a recess portion provided in the holder.

6. The voltage detection device according to claim 5, wherein there is a gap between the socket and the recess portion.

7. The voltage detection device according to claim 5, wherein the plurality of battery cells is stacked in a predetermined first direction, the holder is located on a side of a second direction orthogonal to the first direction with respect to the plurality of battery cells, and the recess portion is provided on a surface of the holder on a side of a third direction orthogonal to the first direction and the second direction.

8. A battery module comprising: the plurality of battery cells; and the voltage detection device according to claim 1.

9. A battery module comprising: the plurality of battery cells; and the voltage detection device according to claim 5.